The Axiom of Finite Representation

When Mathematics Remembers a Symbol is Finite

KEVIN R. HAYLETT

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Euclid's Missing Ink

In about 300 BCE, in the sunlit rooms of Alexandria, a man named Euclid sat down and compiled a textbook he called the *Elements*. For more than two thousand years it became the single most influential work in the history of mathematics, required reading for anyone who wanted to think seriously about numbers, shapes, or proof.

Yet Euclid did not invent the geometry he set down; he inherited it. He gathered definitions from earlier Greek mathematicians, chose the ones that fitted together cleanly, polished them until they gleamed with clarity, and arranged them in an order so logical it still feels inevitable today. His real genius lay not in originality but in the careful and patient work of systematisation.

To do that work, though, Euclid had to make a strange and beautiful decision. He had to forget something essential. Perhaps, let me show you what he forgot, and why remembering it now might change the way we see everything that followed.

The Definitions That Changed the World

Euclid opened the *Elements* with three short statements that would echo through the centuries:

- A point is that which has no part.
- A line is breadthless length.
- The extremities of a line are points.

Perhaps read them slowly. They are not definitions in the modern, airtight sense; a proper definition should build only on terms already understood. Yet here, right at the beginning, Euclid is already using words, “part,” “length,” “extremities,” that he has not yet explained. He cannot explain them yet. He is standing at the very threshold. euclid is claiming the language and beginning the path of stabilising representation and meaning.

If we look closely, we see these are not quite definitions at all. Rather, they are invitations and handshakes with the reader. Euclid is saying, in effect: we both know, in some deep intuitive way, what I mean. Let us agree on these ideal concepts together, and from now on we will speak only of the properties I have listed in the postulates.

> *A point has no parts. A line has no width. A circle is perfect.*

Do such things exist in the world we actually touch? Of course not. A dot of ink has size, shape, and a fuzzy boundary. A line drawn with a pen carries the width of the nib and the way the ink bleeds into the paper. A circle scratched on papyrus is never truly round. It trembles with the tremor of the hand that drew it. Euclid knew this perfectly well and we "know" this. He was no innocent dreamer. He simply chose, with extraordinary clarity, to forget the ink and set the course of mathematics on a journey into the next few thousand years. But perhaps, we need to measure the words carefully, and with same care we build an instrument to measure gravity and the Higgs boson. When we look close we can notice an unstated postulate.

The Unstated Postulate

Let us give a name to what Euclid. Let us call it the unstated postulate that quietly underpins all of classical mathematics:

> *Let it be granted that we ignore and forget the ink.*

That single, unspoken agreement was one of the most fertile fairy tales ever told in the history of thought. By setting aside the physical medium, the wet ink, the stylus, the grain of the surface, the inevitable width of every mark, Euclid stepped cleanly into the realm of pure ideality and imagination.

His points are invisible to the eyes, as they could have no parts because they were no longer physical dots. His lines could be breadthless because they were no longer strokes of any real pen. His circles could be perfect because they were no longer approximations pressed into imperfect material. All would be invisible to the eye and what do we say: mathematics is not real it is an idealised representation.

Yet, here's the thing: we have to write the symbols with real ink that can be measured or instantiate the symbols in memory in silicon. No proof is invisible. Every symbol is measurable and finite even that of infinity. Infinity is a finite symbol. You know this and no doubt you have a flow of text to explain this — but it will be a finite flow of text!

This act of forgetting opened the door to how we see mathematics in terms of a "Platonic certainty", "deduction, and to "proof". All become solid in the mind of the reader. It raised the cathedral of classical mathematics, stone by invisible stone.

However, every forgetting carries a cost. Over two thousand years we began to mistake the ideal for the real. We started to believe that points truly exist somewhere out there, beyond representation, beyond what we can measure. We started to treat the infinite continuum as something “given” rather than something we ourselves had constructed. We came to see the finite as a mere shadow of the ideal, instead of the other way around.

This is not a complaint about Euclid. It is simply what happens when a brilliant story is told so beautifully, and for so long, that we stop noticing it was ever just a beautiful story that we could never measured with our own eyes.

What If We Remember the Ink?

> *Perhaps, now imagine we did the opposite.*

What if we began mathematics not from the ideal but from the ink itself, from the finite, the measured, the uncertain? What if we started where every actual act of thinking actually starts: with a mark on a page, a pixel on a screen, a pattern of neural firing that has size, history, and limits? Ask yourself a simple question:

> ***Is a Symbol Finite?***

For me it seems that when we measure, a symbol is never a Platonic form floating free. It is a physical or computational trace, something that takes up space, that can be measured, that can be told apart from its neighbours only up to a certain resolution. Below that resolution it dissolves into noise. This is not a philosophical aside, rather it is an alternative foundation and leads us to a missing mathematical axiom:

The Axiom of Finite Representation.

> *Every mathematical symbol must be instantiated, physically or computationally realised, in a finite medium that has measurable extent, bounded precision, and a traceable history. No symbol floats free. The representation is the symbol.*

And from it flows a second truth, immediate and inescapable trajectory:

The Corollary of Inescapable Uncertainty.

Every act of mathematics requires measurement, and every measurement carries uncertainty. Even the number 1, written on paper, has an ink spread. Even the cleanest logical deduction, whether performed by brain or silicon, crosses thresholds of signal strength rather than inhabiting infinitely sharp truth values.

What Changes?

Once you let these two ideas settle, many things shift quietly but profoundly.

Equality, for instance, is no longer the same as identity. In classical mathematics two numbers are either equal or they are not; there is no middle ground. But when numbers arise from measurement, two intervals either overlap within the limits of our instruments or they do not. Overlap is the only equality we ever truly possess. It carries a confidence interval rather than a Boolean certainty.

Points, too, lose their absolute status. Nothing physical has "no parts." If mathematics is to speak honestly about the world we actually measure, the smallest geometric object we can locate is not a point but a little sphere, the smallest volume whose centre we can meaningfully identify. At the limit of measurement, uncertainty spreads equally in every direction. The atom of geometry and all we can measure is spherical.

What then of our classical Euclidean "points". Well they remain useful fictions, limits we can approach but never quite reach, and that is fine, so long as we remember they are fictions, they are built from the imagination.

Even proofs change their nature. A classical proof presents itself as an eternal truth, independent of paper, reader, or time. But if every symbol must be instantiated, then a proof becomes a *finite symbolic trajectory*: a path of symbols and words that unfolds on a page, on a screen, or in a mind. Each step is a measurement. Each step carries its own small uncertainty.

The proof remains valid. Only now it is honest about the confidence and the provenance of the checker who walks its path.

The Return of the Ink

Here’s the part most conversations about foundations quietly skip. Classical mathematics presents itself as both ideal and complete: Early in the 20th century commitments were made: real numbers, functions, proofs, all self-contained, cost-free, certain.

Yet the moment we actually use the instrument itself the language of mathematics, building a bridge, simulating a climate, training a neural net, guiding a rocket, the infinite collapses back into the finite. The perfect π becomes a truncated decimal. The exact multiplication becomes a rounded operation in finite registers. The timeless proof becomes a sequence of steps that each take time, consume energy, and live in bounded memory. The ink returns, the cost returns, and the uncertainty returns.

This is not a failure of classical mathematics. Rather, it is the inevitable signature of any instantiated symbolic system. Classical formalism is the elegant bridge that sits between two inescapably finite acts: the act of measuring the world to produce a symbol, and the act of using one set of functional symbolic trajectories to create another. Classical mathematics, lives in the endogenous world, abstracting away both ends, pretending for a while that it needs nothing but itself.

This is where my explorations and work start in what I call Finite Symbolic Mechanics. Rather than reject the ideal it simply refuses to confuse the ideal basin with the one where measurements are the foundation of the symbols we use. For me, Finite Symbolic Mechanics supplies the ground that classical mathematics abstracts away, and then, the moment it is put to work, must return to.

So In FSM we have the ability to see stable sequences of words, as functional symbolic trajectories that we can map into semantic phase space. The stability comes from the wider space of language. Now with this in mind we can examine how Euclid went from measurements to the imagined point. Because in the more detailed earlier texts of Euclid this made clear.

The Geometer’s Point

While writing this essay I was fortunate enough to acquire a 1607 edition of Euclid’s Elements with the commentary of Christoph Clavius. Reading the original Latin commentary led me to notice something that is often hidden in modern presentations.

Today we are usually taught that a point is simply an exact location. We draw a dot, place it on a Cartesian grid, assign coordinates, and continue. The point appears as though it were already there waiting to be found.

The older commentary proceeds differently. Before speaking of the point, Clavius discusses measurable extension itself; he speaks of length, breadth, and depth. Some magnitudes possess one dimension, some two, some three. These are things we can recognise in experience, things we can draw, compare, and measure.

Then the reader is asked to perform an unusual act.

- Remove depth.
- Remove breadth.
- Remove length.
- Remove every divisible part.
- Remove every material example.

The commentary even notes that the tip of a needle fails as an example, because it remains something that could still be divided.

Only after this process of subtraction does the point appear. The point is not discovered, it is not measured, its generated and imagined into being.

Stated cleanly, we can see the geometer is asked to hold an object that possesses no length, no breadth, no depth, and no material existence. The point is admitted not to exist among physical things, yet the geometer proceeds as though it does.

This is not a criticism of Euclid; quite the opposite as it reveals the extraordinary achievement at the heart of mathematics. Euclid was not merely defining objects. Rather, he was teaching his readers how to stabilise acts of imagination and how to think, even if it was to think of invisible clothes.

The point is therefore not a mark on paper. Every mark possesses extent, contains ink, and can be measured. Instead, the point is a shared symbolic construction and an artifice. Language leads the reader from measured experience into an admissible imagined object that can then be used to build the rest of geometry.

> *Seen this way, the Euclidean point is born from measurement but does not remain within measurement.*

The point begins as length, breadth, depth, division, and material experience. Then, we these measurements are progressively stripped away until only an indivisible symbolic limit remains. The reader is not shown the point, but taught how to see as a geometer.

For me, this distinction matters as the moment we wish to calculate, draw, prove, or compute, the point must once again acquire clothing. We give the point a letter such as A, assign coordinates. We then write the symbol down with ink or store it in memory and render it as pixels. We represent it through finite symbols.

The ideal geometers point has no parts, yet the operational point always has representation. In that sense, Euclid first removes the clothes from the point so that geometry can begin. Computation then puts the clothes back on so that geometry can be used.

The point is perhaps the first great example of mathematics transforming measured experience into a stable act of imagination and then returning that imagination to the world through finite representation. The stabilisation is a social act of consensus. This is part of the arc of commitment, rules of admissibility, and stabilisation. We collectively stabilise the imagined point through a commitment to infinity and shared rules of admissibility and use.

The Pebble and the Cairn

After taking my artistic licence out of my wallet perhaps let me tell you a small myth I made up, though it feels less like invention and more like remembering something very old. The first word was not “god,” not “mother,” not even “one.” The first word was **pebble**.

A pebble is small. It has edges, can be held in the hand and compared to another pebble: larger, smaller, smoother, sharper. It carries provenance: it came from a particular riverbed, a particular mountainside. It can be carried, given, arranged, built with.

Every later word, every abstraction, every slow noun and verb, is a distillation of that first concrete act. One pebble is a symbol. Two pebbles make a relationship and three begin to form a pattern. Stack them carefully, one upon the last, and you have a cairn, a visible trajectory that says: this way, I was here, and follow me to the cairn.

> *Mathematics is a cairn. A proof is a cairn. A conversation is a cairn.*

Euclid directed attention away from the physical marks and toward their idealized relationships. This is where my own work starts to explore, it's the world of measure, ad for name I call the work Finite Symbolic Mechanics. And it explores a world where the inks is accounted for within the symbols we use. And we hold measurement to be the foundation and symbols have to be measured, and they have preveiance and uncertainty. they are all we have, and it seems to me they are more than enough.

What About Classical Mathematics?

Classical mathematics remains extraordinarily useful. The real numbers, the continuum, the infinite, these are powerful tools in the realm of mathematical theory. The theories point to to the real measured calculations that need finite numbers that let us calculate orbits, design bridges, model climates, and build the very computers on which we argue about foundations.

However, they are perhaps best seen as the limit case of finite mathematics as uncertainty tends to zero, as extent tends to zero, as provenance is politely ignored. They are what you get when you approach the limit of actual measurements and calculations. The region of representation were numbers have to be measurable and carry uncertainty.

So, we do not need to burn the wonderful cathedrals of classical mathematics. We only need to remember the ground they stand on as the ideal is downstream of the real, not the other way around. To do this, perhaps, we need to include the missing axiom.

Why This Matters Now

It seems to me that none of this is abstract philosophy anymore. The more we enter a world that embedded in computation, the further we move from the act of writing and seeing the ink on the page. Information becomes more abstract. It matters first for artificial intelligence as large language models generate symbols one token at a time. They have no access to the infinite, no real numbers, only finite weights measured during training. For example, when an AI "hallucinates," it is not breaking; it is simply drifting off the intended trajectory while remaining locally fluent. A mathematics of finite trajectories helps us understand that.

And then it matters for measurement itself as modern science rests on precise metrological acts. The SI second is defined by assigning it to 9,192,631,770 periods of caesium radiation, an assignation by committee. That is not an abstract definition; it is a maintained symbolic pathway anchored in instruments, institutions, and repeated human practice. To discuss the success of mathematics we must not forget where the measurements start and the chain of provenance that create the stability of the numbers we use.

Importantly, it matters because we now live in an age where we can measure and acknowledge the limits: finite computing power, finite energy, finite attention. And we all know know, deep down, that "infinite" precision, infinite sets, the completed continuum all hide the real costs of instantiation and can never be measured. Now, more than ever, mathematics that makes those costs visible is no longer optional; it is necessary, because weed to find the stability that holds meaning and representation.

The Call

Euclid forgot the ink and perhaps that forgetting was brilliant. It bought us two thousand years of luminous certainty. But we are no longer required to live only in that basin, we can remember the finite nature of representation, that is to say the ink.

We can begin to build mathematics on finite representation, inescapable uncertainty, and traced provenance. We can treat proofs as functional symbolic trajectories, numbers as measurements, and points as the useful fictions they have always been. This is not a rejection of Euclid but perhaps the trajectory that is necessary, the quiet act of picking up the pebble he set down and beginning the next layer of the cairn. The basin is of the finite age is open, and the finite symbolic trajectory is still unfolding so let us build.

A Note on the Ink

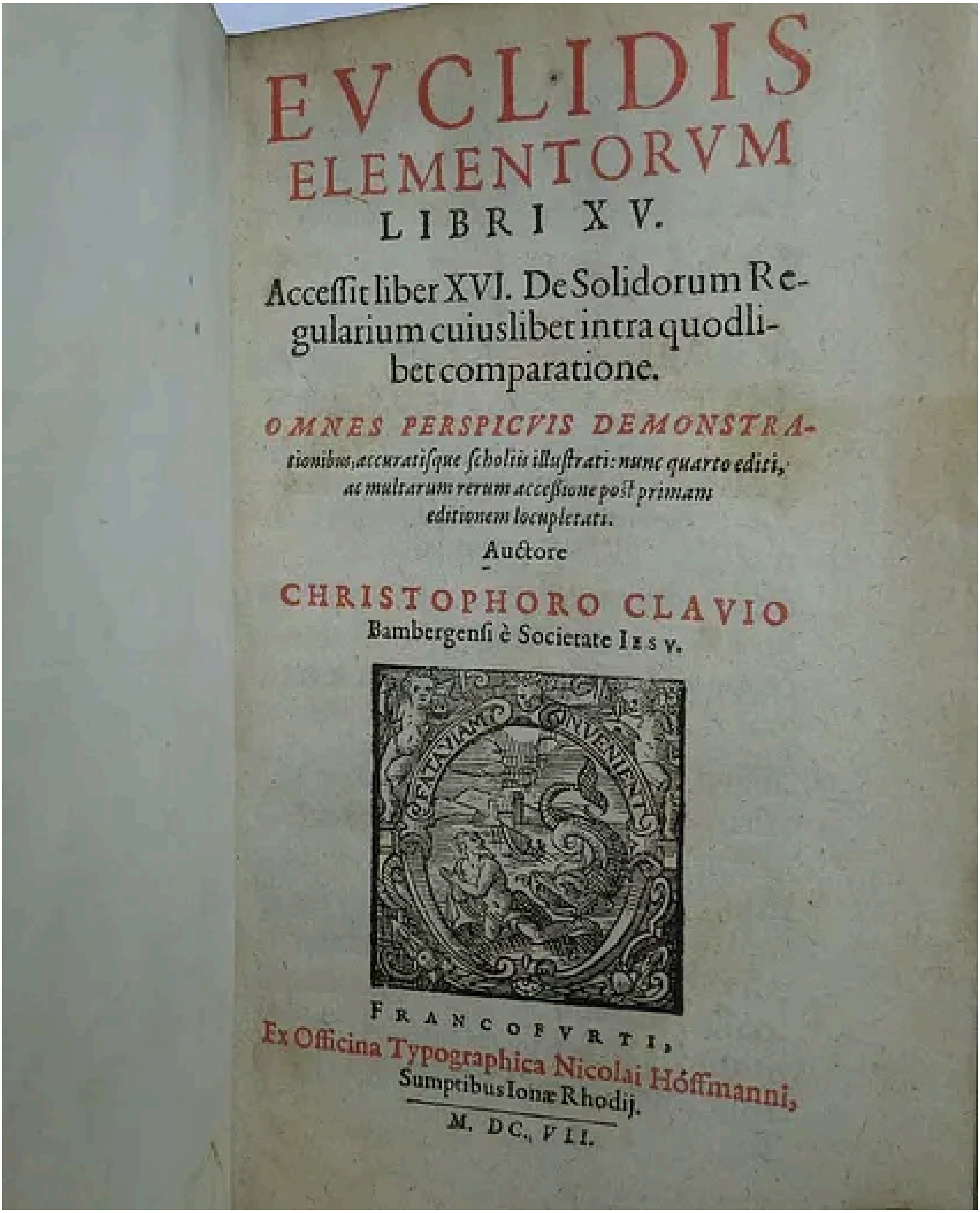
The essay you have just read spoke often of Euclid forgetting the ink. That phrase is not a metaphor. There's a book on my bookshelf: it is Christoph Clavius's edition of Euclid's *Elements*, printed in Frankfurt in 1607, six years before Galileo turned his telescope toward the moons of Jupiter, the same year Jamestown was founded in Virginia.

The book's vellum binding is animal skin that still breathes and warps with the weather. The type is impressed into laid paper; over four centuries the ink has gently spread into the fibres. Some margins carry faint annotations in an old hand, a student's note, a scholar's query, perhaps even a prayer. A single blank page has been carefully excised; we will never know why.

Euclid forgot the ink, but the ink never forgot Euclid. Every point in this book has a tiny smudge and every line has width. Every circle is a woodcut, cut by hand, printed by pressure, imperfect. Yet once instantiated by finite marks on paper the geometry works, the results hold when computed with finite numbers. The theorems hold well enough, the *Elements* remains readable, teachable, and finitely true.

That quiet miracle of how finite mathematics works well enough is what my work in Finite Symbolic Mechanics tries to name. It does this with uncertainty, provenance, and with finite representation. Yet, the ideal and the real are not enemies; rather they are partners. The ideal is what we can say when we forget the ink, the measured is what we hold when we remember we have to measure with our eyes and instruments and commit the measurements to a mark or in a computer's memory.

Euclid's Elements is still legible after 419 years. And, yes, it is a story of ideals, however it needs the finite marks on the paper to carry the work forward so we can measure it today and as the future unfolds into new trajectories.



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Sumptibus Ionæ Rhodij.
M. DC. VII.

Francofurti, MDCVII. The ink is still legible after 419 years. The words within are a trajectory and not a monument.

Further Reading

Monograph on Functional Symbolic Mechanics

Essay on Functional Symbolic Trajectories

Finite Tractus: The Hidden Geometry of Language and Thought

Pairwise Phase Space Embedding in Transformer Architecture

Introducing the Takens-Based Transformer

JPEG Compression of LLM Input Embeddings

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Further Papers, Monographs, and Essays

About the author

Earlier works from a career in Biomedical Engineering

Note: most articles are posted on both Substack and Medium so you can choose your preferred format for reading.

> *Omne quod est, finitum est; tantum per mensuram cognosci potest*
>
> *Everything that exists is finite; it can only be known by measure*
>
> *kevinhaylett.substack.com* | *geofinitism.com*
>
>

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